

Lloyd Akresi

lloydakresi@gmail.com | Takoradi, Ghana | GitHub

Education

BSc. Electrical/Electronic Engineering Kwame Nkrumah University of Science and Technology, Kumasi, Ghana GPA: 79.73/100 Advisor - Dr. Charles M. Adrah Capstone - Machine Learning for Fiber Sensing Using State of Polarization Awards - Provost's List of Excellent Students (Two -Time Recipient - 2023, 2024)	Jan. 2022 – Nov. 2025
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Research Interests

I am interested in interpretable and robust machine learning, particularly how model behavior can be made trustworthy for high stakes decisions rather than treated as a black box. My toxicity classification project addresses this through robustness under severe class imbalance, and I am drawn to work on embedding fairness and alignment constraints directly into learning objectives. My physics informed neural network for DC motor state estimation and subsea anomaly detection pipeline extend machine learning into industrial and subsea infrastructure, domains where I am interested in exploring what interpretability and trust actually require.

Research Experience

Undergraduate Research Assistant <i>Power Systems Laboratory, KNUST</i>	Jan 2024 – Dec 2024 Kumasi, Ghana
◦ Benchmarked Modified Artificial Rabbit Optimization (MARO) against PSO, ABC, ARO, and RIME across 10 benchmark functions, contributing to a peer-reviewed publication on distribution network reconfiguration. ◦ Validated algorithm performance on the IEEE 33-bus system, where MARO achieved the lowest power loss (0.144 MW) and best voltage profile among all compared methods. ◦ Demonstrated MARO's superiority over PSO, ABC, ARO, and RIME across four statistical metrics (optimum value, mean, STD, MAE), establishing it as the top-ranked algorithm overall.	

Professional Experience

TechnipFMC <i>Technical Services Trainee</i>	Nov. 2025 – Present Takoradi, Ghana
◦ Supported 20+ high-pressure testing operations (up to 10,000 psi), including subsea equipment pressure tests and fluid pre-charge scopes, with responsibility for instrumentation monitoring, leak detection, and accurate test documentation ◦ Verified electrical and hydraulic control system functionality of subsea control modules using proprietary diagnostic software, ensuring system integrity prior to offshore deployment ◦ Calibrated instrumentation (LVDTs) and applied Modbus and CANbus protocols to test and validate chemical injection metering valves and acoustic sand detectors, improving reliability of instrumentation systems	
Volta River Authority <i>Electrical Engineering Intern</i>	Sept. 2024 – Nov. 2024 Tema, Ghana
◦ Conducted preventive and corrective maintenance on balance-of-plant equipment to optimize system performance. ◦ Gained hands-on experience in power generation processes and critical machinery functions. ◦ Assisted engineers in troubleshooting steam turbine actuator systems, supporting fault diagnosis and inspection of control components within turbine control mechanisms	
Volta River Authority <i>Electrical Maintenance Intern</i>	Aug. 2023 – Nov. 2023 Takoradi, Ghana
◦ Conducted electrical testing on power system components including transformers, generators, and earthing systems, supporting verification of insulation integrity, grounding effectiveness, and operational safety ◦ Assembled and tested Direct-On-Line (DOL) motor starter circuits, verifying correct wiring, contactor operation, and protection functionality for safe motor startup	

Publications

Twumasi, E., Dogbey, F. K., Teye, F., **Akresi, L. N.**, Souto, L., Frimpong, E. A. "Distribution Network Reconfiguration Using a Modified Artificial Rabbit Optimization Algorithm," 2025 IEEE PES/IAS PowerAfrica Conference, Cairo, Egypt, 2025.[\[Link\]](#) [↗](#)

Projects

- Ragify** [Github](#) [↗](#)
Document-level RAG system using LangChain and Sentence Transformers for ingestion, ChromaDB retrieval with cross-encoder reranking, session management with deduplication and TTL/LRU eviction, and served via Gradio and Groq with Docker deployment.
- ArXiv Lens** [HuggingFace Spaces](#) [↗](#)
Semantic search engine over 50k CS/ML ArXiv papers using ChromaDB, Sentence Transformers, CrossEncoder reranking, and Groq API, deployed on HuggingFace Spaces.
- Toxic Comment Moderation API** [Github](#) [↗](#)
Multilabel toxicity classifier fine-tuned on the Jigsaw dataset using DistilBERT, with class imbalance handling via weighted BCE loss, served as a REST API with FastAPI.
- DC Motor State Estimation via Physics-Informed Neural Network**
Physics-informed neural network for real-time state estimation of a DC motor, embedding governing differential equations directly into the loss function to enforce physical constraints.
- Subsea Anomaly Detection Pipeline**
Hybrid autoencoder and Gaussian Mixture Model pipeline for explainable anomaly detection in subsea fiber-optic recordings, developed as an undergraduate thesis project.
- Fault Detection via Temporal Fusion Transformer** [Github](#) [↗](#)
Transformer-based explainable fault detection pipeline on the Tennessee Eastman Process dataset using a Temporal Fusion Transformer for multivariate time-series anomaly classification.
- Machine Language Translator** [Github](#) [↗](#)
Full Transformer architecture implemented in PyTorch without `nn.Module`, including multi-head attention, positional encoding, decoder masking, and LayerNorm for translating English to French.

Relevant Coursework

C, MATLAB, Classical and Digital Control Systems, Analog and Digital Communication Systems, Digital Signal Processing, Computer Networking

Technical Skills

- Programming:** C, Python, JavaScript, Ruby, L^AT_EX
Libraries: PyTorch, NumPy, Scikit-learn, HuggingFace Transformers, Pandas, FastAPI, Docker, Polars
ML & Data: ChromaDB, Sentence Transformers, Weights & Biases, Gradio, Streamlit
Industrial Control: Siemens TIA Portal, WinCC, LogixPro
Networks: GNS3, Cisco IOS
Developer Tools: Git, GitHub, Linux, VS Code, Jupyter

Membership & Service

General Secretary , KNUST Electrical/Electronic Engineering Students Association	Aug. 2023 – Aug. 2024
General Secretary and Publicity Head , KNUST IEEE Instrumentation and Measurement Society	Jul. 2024 – Jun. 2025
Editorial Team Member , TechAble	Jan. 2025 – May 2025
Member , IEEE Computer Society	Jan. 2025 – Present
Member , IEEE Power Electronics Society	Jan. 2025 – Present
Student Member , Ghana Institute of Engineers	Sept. 2025 – Present
Mentor , KNUST Electrical/Electronic Engineering Students Association	Jan. 2023 - Sept. 2025